Supplementary Material
Frozen Few-Shot Class-Incremental Learning is a Tri-Level Calibration Problem

BMVC 2026 Submission # 1525

1 Empirical Observations Behind the Tri-Level Reformulation

Fig. 1 quantifies the two structural observations made in the introduction of the main paper. Panel (a): the mean off-diagonal cosine similarity of visual prototypes in the final session shows CLIP prototypes are visually dense across all three benchmarks while DINOv2 prototypes are an order of magnitude sparser; the two encoders therefore cannot be averaged in embedding space. Panel (b): the Pearson correlation between per-class final-session accuracy and the session in which the class was introduced is moderately negative on CIFAR-100 and *mini*ImageNet but approximately zero on CUB-200, indicating that on fine-grained data the prototype noise tracks local cluster density rather than session age.

2 Session-wise Comparison Tables

Tables 1 and 2 give the per-session breakdowns on CIFAR-100 and CUB-200 corresponding to Table 1 of the main paper.

3 Robustness Ablations

Cross-dataset fixed-α robustness. Table 3 reports the 20-seed final-session Full accuracy of TRIC-QPR across $\alpha \in \{0.5, 0.75, 1.0\}$ on all three benchmarks. The variation is at most 0.25 points on CIFAR-100, 0.10 points on *mini*ImageNet, and 0.10 points on CUB-200, so the default $\alpha = 0.75$ is consistently near the local optimum and not dataset-specific.

Alternative query-update rules. The gain of QPR is not only an effect of using the query batch, but of using it conservatively. Fig. 2 compares three available query-side novel-prototype updates on the BiMC ensemble backbone over five matched seeds per dataset: our conservative gate, a hard pseudo-label update with the same minimum-count constraint, and an unconstrained transductive update. The conservative gate is the safest option overall: it matches or exceeds hard pseudo-labeling in Full accuracy, preserves base classes better on CUB-200, and avoids the large degradations of the transductive update.

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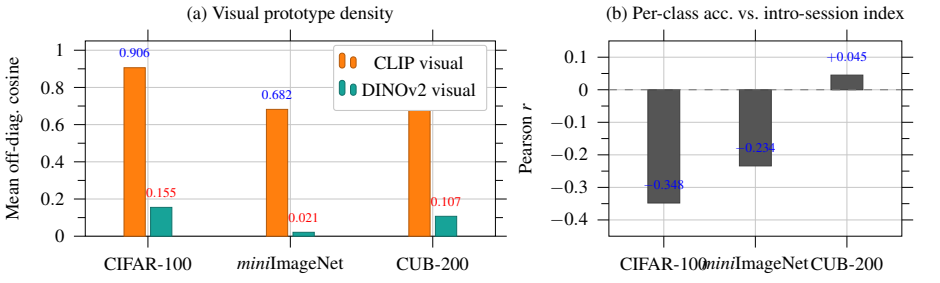


Figure 1: Quantitative observations underlying the tri-level reformulation. (a) CLIP visual prototypes are visually dense, DINOv2 prototypes are an order of magnitude sparser. (b) Per-class final-session accuracy correlates negatively with the introduction session on CIFAR-100 and *mini*ImageNet but is approximately uncorrelated on CUB-200.

Method	Modality	Accuracy in each session (%) ↑									Avg. ↑	PD ↓
		A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8		
CLIP zero-shot	L	74.77	73.12	72.56	71.08	70.46	70.41	70.58	70.06	68.93	71.33	5.84
Visual prototype (CLIP)	V	75.88	72.78	71.01	68.52	66.55	64.84	64.12	62.71	61.03	67.49	14.85
TEEN	V	75.88	73.00	71.39	68.88	67.14	65.41	64.94	63.44	61.84	67.99	14.04
FeCAM	V	81.38	78.28	76.93	74.55	72.91	71.81	71.26	69.80	68.32	73.92	13.06
BiMC	V-L	79.70	77.97	77.29	75.35	74.76	74.39	74.34	73.55	72.50	75.54	7.20
BiMC Ens.	V-L	81.98	79.74	78.84	77.00	76.11	75.68	75.33	74.61	73.18	76.94	8.80
CLIP visual only	V	75.90	72.74	70.97	68.31	66.54	65.49	64.94	63.73	62.20	67.87	13.70
DINO visual only	V	89.63	87.87	87.21	85.16	85.22	84.90	84.92	84.57	83.34	85.87	6.29
TRIC	V-L	90.90	89.37	88.87	87.15	87.22	86.95	87.08	86.81	85.78	87.79	5.12
TRIC-QPR	V-L	90.90	89.78	89.42	88.04	88.17	87.98	88.12	87.91	87.13	88.61	3.77

Table 1: Session-wise comparison on CIFAR-100. V, L, and V-L denote visual, language, and vision-language modalities; A_t is accuracy at session t, Avg. is mean session accuracy, PD is base-to-final degradation.

4 Prediction Entropy

Table 4 reports BiMC reference prediction entropy in the final session, separately for base and novel classes, across textual, visual, and calibrated classifiers. The calibrated classifier is the lowest-entropy variant on every dataset, which together with the complementarity analysis in the main paper (Fig. 3) confirms that the increased confidence tracks correctness.

5 Per-Class Behavior of TRIC-QPR

A fixed-seed, per-class diagnostic traces where each gain arises and where the current design is conservative.

The gains are sparse and concentrated. On CIFAR-100, 39 classes improve, 20 get worse, and 41 are unchanged in the final session; the splits are 19/11/70 on *mini*ImageNet and 30/13/157 on CUB-200. Aggregate gains are carried by a minority of novel classes whose five-shot prototypes were far from the local cluster centroid; most classes are untouched. This pattern is consistent with the conservative geometry of QPR and indicates that the current uniform α leaves potential gains on the table for classes with abundant high-purity gated queries.

Method	Modality	Accuracy in each session (%) ↑											Avg. ↑	PD ↓
		A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	A_9	A_{10}		
CLIP zero-shot	L	66.38	64.05	62.71	59.07	59.24	59.27	57.76	56.69	55.46	55.19	55.73	59.23	10.65
Visual prototype (CLIP)	V	81.42	78.78	77.43	74.22	72.91	71.59	70.65	69.97	68.79	68.93	68.80	73.04	12.62
TEEN	V	81.42	79.16	77.49	74.60	73.26	71.75	70.68	70.21	68.90	69.28	69.12	73.26	12.30
FeCAM	V	82.26	79.48	77.76	74.68	72.86	71.10	69.92	69.17	67.61	67.89	67.40	72.74	14.86
BiMC	V-L	82.16	79.99	79.04	76.10	75.07	74.04	73.13	73.08	71.88	72.26	72.25	75.36	9.91
BiMC Ens.	V-L	83.00	81.01	79.83	76.96	75.84	74.73	73.86	73.47	72.26	72.61	72.68	76.02	10.32
CLIP visual only	V	81.63	79.14	77.49	74.24	73.26	71.68	70.81	70.03	68.53	68.72	68.47	73.09	13.16
DINO visual only	V	86.35	85.19	84.66	83.67	83.68	82.34	82.72	83.28	83.14	83.65	84.01	83.88	2.34
TRIC	V-L	87.64	86.25	85.80	84.83	84.91	83.74	84.10	84.59	84.35	84.82	85.17	85.12	2.47
TRIC-QPR	V-L	87.64	86.58	86.23	85.40	85.66	84.46	84.84	85.32	85.16	85.62	85.97	85.72	1.67

Table 2: Session-wise comparison on CUB-200-2011. V, L, and V-L denote visual, language, and vision-language modalities; A_t is accuracy at session t, Avg. is mean session accuracy, PD is base-to-final degradation. The final session column is A_{10}.

α	CIFAR-100	*mini*ImageNet	CUB-200
0.50	85.42	94.68	86.12
0.75 (default)	85.50	94.74	86.17
1.00	85.25	94.78	86.08

Table 3: Cross-dataset robustness of the query-mean interpolation weight α. Values are 20-seed final-session Full accuracy of TRIC-QPR.

The gate is clean, but intra-novel contamination can still occur. Final-session base leakage is 2.39% on CIFAR-100, 1.16% on *mini*ImageNet, and 0.63% on CUB-200. Inside the admitted-novel set, however, queries can still be assigned to the wrong novel class. On CUB the `Caspian Tern` class drops from 50.0 to 16.7 because the admitted pool is contaminated by visually similar tern queries whose calibrated argmax is `Elegant Tern`; `Caspian Tern`'s prototype then starves. Reporting only "predicted-class-is-novel" purity hides this failure mode, motivating an *intra-novel contamination* metric as a standard diagnostic and a gate that conditions on the second-best novel competitor.

The refined DINOv2 geometry changes locally, not globally. The mean off-diagonal DINOv2 similarity changes by only $+0.006$, $+0.001$, and $+0.003$ across datasets after QPR. The refinement relocates a small number of novel centroids within their local fine-grained neighborhood rather than reshaping the global metric, which is why QPR delivers a positive novel-class gain on CUB-200 even though Fig. 1(b) shows its noise source is local density rather than session age.

6 Limitations and Concrete Extensions

A structural gap between support and query. TRIC uses the labeled support; QPR uses the unlabeled query batch. Between them, there is no *labeled-only* refinement of the novel DINOv2 prototype itself. BiMC's soft visual calibration filled this slot for CLIP visual prototypes by pulling novel centers toward base classes via attention; we removed it for DINOv2 because the cross-class attention pulls novel centers off their true cluster. A clean replacement would be labeled-only and intra-novel: a leave-one-out support refinement that uses the remaining shots of class c to denoise the centroid, or a margin-aware contrast that pushes novel prototypes away from their dominant DINOv2 neighbors.

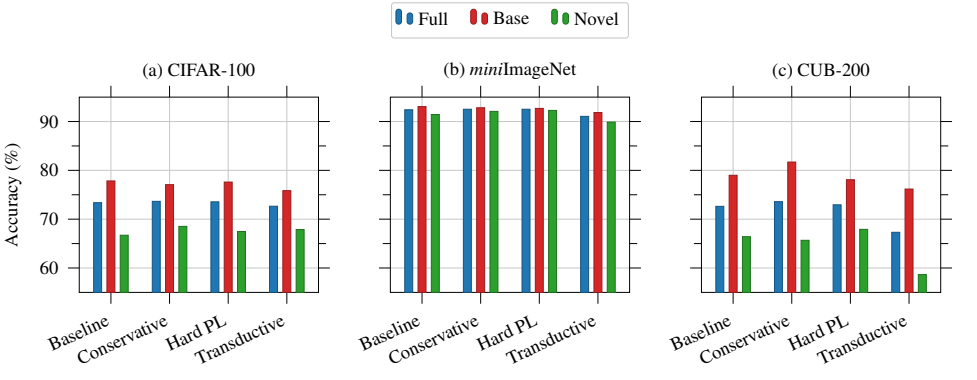


Figure 2: Alternative query-update rules on the BiMC ensemble backbone. Bars show final-session Full / Base / Novel mean accuracy over five matched seeds. The conservative gate is the most stable rule overall: hard pseudo-labeling trades away more base accuracy for novel gain, while the transductive update collapses both base and novel accuracy — especially on the fine-grained CUB-200 benchmark.

Classifier	CIFAR-100		*mini*ImageNet		CUB-200	
	Base entropy ↓	Novel entropy ↓	Base entropy ↓	Novel entropy ↓	Base entropy ↓	Novel entropy ↓
BiMC textual classifier	1.798	1.933	0.481	0.554	1.297	1.778
BiMC visual classifier	1.619	2.036	0.235	0.228	1.141	1.810
BiMC calibrated classifier	1.199	1.395	0.174	0.178	0.951	1.511

Table 4: BiMC reference prediction entropy in the final session. Values are reported separately for base and novel classes.

Uniform α is geometrically conservative. A single α across all novel classes treats every class as equally trustworthy, which contradicts the fine-grained density variation we observe. A per-class α informed by gated-pool size, local prototype density, or posterior confidence is a natural extension.

Per-dataset inter-model weight ω. We fix $\omega = 0.7$ across datasets. On CIFAR-100 the isolated DINOv2 visual branch already exceeds the fused visual branch before QPR, while on *mini*ImageNet and CUB-200 the CLIP visual contribution remains useful. A per-dataset ω chosen by validation would address this without compromising the label-free deployment claim.

Intra-novel gate. The current "predicted-class-is-novel" gate purity does not catch intra-novel contamination (the `Caspian Tern` case above). A gate that additionally rejects queries whose top-two novel posteriors are close would close this leak.